



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

AGENTIC AI - 22AIM813

QUESTION BANK

MODULE 1

1. Comparison between traditional AI, Generative AI and Agentic AI with examples
2. Explain the architecture and working of an Agentic AI system with a suitable example.
3. Discuss autonomy, goal-driven behavior, and multi-step reasoning.
4. Architecture and working of an Agentic AI system
5. Define intelligent agent. What is decision-making in Agentic AI?
6. Illustrate working of Agentic AI
7. Explain the concept of Agentic AI, its key characteristics and working with an example.
8. Write the concept of autonomy in intelligent agents.
9. List the use of agents in Autonomous patient monitoring and treatment recommendation systems.
10. Discuss the role of Agentic AI in autonomous vehicles?
11. Write the various functionalities of Traditional AI, Generative AI and Agentic AI
12. List the following features of Agentic AI:
(i) Multi-step reasoning (ii) Goal-driven behavior
13. Explain the core components of Agentic AI with a neat diagram.
14. Explain Agentic Reasoning. How do Multi-step Reasoning Agents Work?
15. Explain the key features of Goal-Driven Agents?
15. Discuss the role of Large Language Models (LLMs) in Agentic AI.
16. Analyze the benefits and challenges of deploying Agentic AI in real-world systems.

MODULE 2

1. Apply prompt engineering techniques to design an effective prompt for a task like summarization or question answering.
2. Analyze how attention mechanisms enable context awareness in large language models.
3. Working of transformer? Importance of encoder and decoder?
4. Examine ethical issues in LLM usage such as bias, misinformation, and data privacy, and suggest mitigation strategies.
5. Explain different prompt engineering techniques.
6. Evaluate the role of LLMs in building intelligent AI agents with real-world applications.
7. What is self-attention?
8. Illustrate the key components of the Transformer model with a neat diagram.
9. Write about self-consistency and a tree of thoughts prompting techniques to improve the reasoning accuracy.
10. List the ethical challenges in deploying LLMs, including bias, misinformation, and privacy concerns.
11. Illustrate the role of LLM in building AI agents with a neat diagram.
12. Describe how prompt engineering improves the performance of LLMs in complex tasks.
13. Write the concept of Attention mechanism in capturing dependencies of input features.
14. Explain zero-shot and few-shot prompting with examples.
15. Describe the attention mechanism used in Transformers. Explain how self-attention and multi-head attention improve model performance.
16. Discuss prompt engineering in Large Language Models. Explain basic and advanced prompting techniques with suitable examples.
17. Discuss the challenges and limitations of Large Language Models

MODULE 3

1. Explain the concept of LangChain.
2. Explain what is meant by Chains in LangChain.
3. Explain the role of Tools in LangChain.
4. Explain the concept of Memory in LangChain.
5. Explain what an Agent is in LangChain.
6. Explain the ReAct pattern (Reasoning + Acting) and analyze how it improves agent performance.
7. Explain LangChain workflow?
8. Use the ReAct framework to improve task execution in AI systems?
9. Discuss the use of Tools in Lang Chain to enable an AI system to interact with external APIs or databases.

10. Summarize the concept of Memory and its types in Lang Chain.
11. Illustrate the architecture of LangChain, highlighting the roles of Chains, Tools, Memory, and Agents.
12. Discuss the design challenges in building agentic systems, including memory management, tool selection, and reliability.
13. Summarize the role of prompt engineering in controlling agent behavior within LangChain systems.

MODULE 4

- 1.Explain LangGraph and its role in agent orchestration.
 - 2.Explain task decomposition in AI planning agents.
 - 3.Explain what AutoGPT-style agents are.
 - 4.Explain the concept of reflection loops in AI agents.
 5. Explain self-evaluation in AI agents.
 - 6.Explain the role of graphs in agent orchestration and what are the challenges.
 - 7.Explain Constitutional AI principles in simple terms.
 - 8.Explain how task decomposition helps in solving complex goals.
 - 9.Explain how reflection improves reliability and accuracy in AI agents.
 - 10.Explain why self-correction is important in autonomous AI agents.
 11. Characterize the concept of graph-based agent orchestration in LangGraph and differentiate it from linear chain-based execution
 12. Illustrate the task decomposition in planning agents of AutoGPT-style.
 13. Draw the concept of reflection loops in AI agents.
 14. Investigate the workflow using LangGraph that includes task decomposition, execution, and reflection.
 15. Categorize the self-evaluation and self-correction mechanisms in agent systems.
- Illustrate the principles of Constitutional AI. How does it guide safe and ethical AI behavior?
- 16.Describe how nodes and edges function in a LangGraph for task management.
 17. Describe the working of AutoGPT-style agents in automated planning.

MODULE 5

- 1.Explain FAISS and its role in AI systems.
- 2.Explain tool calling and function calling in multi agent systems
- 3.Illustrate AutoGen and CrewAI in the context of multi-agent collaboration
- 4.Analyze the advantages of RAG over traditional LLM-based approaches in information retrieval tasks.
- 5.Compare single-agent vs multi-agent systems in terms of efficiency, scalability, and reliability.
- 6.Explain Pinecone and its use cases.

7. Explain Chroma and its advantages.
8. Evaluate the concept of Retrieval-Augmented Generation (RAG) to enhance the performance of intelligent agents?
9. Illustrate the role of FAISS vector database in similarity search.
10. Draw the tool calling and function calling mechanisms in LLMs to enhance agent capabilities?
11. Examine the concept of multi-agent collaboration systems using CrewAI.
12. Illustrate the working of chroma and its advantages in vector storage.
13. Use AutoGen in building next-gen LLM applications based on multi-agent conversations.
14. Explain FAISS indexing techniques (Flat, IVF, HNSW) and their impact on retrieval Performance.
15. What is a vector embedding and why is it important in RAG?